

Amelia E. Barber, PhD

Professor and Group Leader

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<https://barber-lab.com>

Research Interests

Phylo- and population genomics of fungal pathogens
Antifungal resistance
Evolution and pathoecology of microbial virulence

Education

- 2016 **PhD**, Microbiology and Immunology
University of Utah, Salt Lake City, UT (USA)
"Zebrafish reveal variable effects of Escherichia coli during sepsis"
- 2006 **BS**, Medical Microbiology & Immunology; Women's Studies
University of Wisconsin-Madison, Madison, WI (USA)

Academic Positions

- 2026 - present **Group Leader**, Experimental Mycology and Microbiome Research
Research Center Borstel-Leibniz Lung Center, Borstel, Germany
Professor (W3), Institute of Experimental Medicine
Kiel University, Kiel, Germany
- 2022 - 2026 **Independent Junior Group Leader.**
Excellence Cluster Balance of the Microverse
Institute for Microbiology, Friedrich Schiller University, Jena, Germany.
- 2020 - 2022 **Junior Group Leader.** Integrated into the group of Gianni Panagiotou.
Leibniz-HKI, Jena, Germany.
- 2016 - 2020 **Postdoctoral Fellow**, Research group of Oliver Kurzai.
Leibniz-HKI and University of Würzburg, Jena and Würzburg, Germany.
- 2010 - 2016 **Doctoral Researcher**, Research group of Matthew A. Mulvey.
University of Utah, Salt Lake City, UT, USA.
- 2007 - 2009 **Associate Research Specialist**, Research group of Clive N. Svendsen.
University of Wisconsin-Madison, Madison, WI, USA.

Selected Publications

Hatmaker AE, **Barber AE**, Drott MT, Sauters TJC, Gumilang A, Alastruey-Izquierdo A, Garcia-Hermoso D, Eagen JL, Keller NP, Kontoyiannis DP, Kurzai O, Rokas A. Evidence for a link between pathogenicity and population structure in a fungal pathogen of humans. *Nature Communications*. 2025.

Campos Rocha M, Lerer V, Adeoye J, Haybe H, Fabre ML, **Barber AE**, Shlezinger N. Aspergillus dsRNA virus drives fungal fitness and pathogenicity in the mammalian host. *Nature Microbiology*. 2025.

Brassington PJT, Klefisch FR, Graf B, Pfüller R, Kurzai O, Walther G, **Barber AE***. Genomic reconstruction of an azole-resistant Candida parapsilosis outbreak and the creation of a multilocus sequence typing scheme. *The Lancet Microbe*. 2024. *corresponding author

Loos D⁺, Pereira A⁺, Dutilh B, **Barber AE**^{*}, Panagiotou G^{*}. A global survey of host, aquatic, and soil microbiomes reveals shared abundance and genomic features between bacterial and fungal generalists. *Cell Reports*. 2024. ^{*}equal contribution, ^{*}corresponding authors

Mirhakkak MH, Chen X, Sae-Ong T, Xu L, Heinekamp T, Kurzai O, **Barber AE**, Brakhage AA, Boutin S, Schäuble S, Panagiotou G. Genome-scale metabolic modeling of *Aspergillus fumigatus* strains reveals growth dependencies on the lung microbiome. *Nature Communications*. 2023.

Barber AE, Sae-Ong T, Kang K, Li J, Walther G, Panagiotou G, Kurzai O. *Aspergillus fumigatus* pan-genome analysis identifies genetic variants associated with human infection. *Nature Microbiology*. 2021.

Barber AE^{*}, Riedel J, Sae-Ong T, Kang K, Brabetz W, Panagiotou G, Kurzai O^{*}. Effects of Agricultural Fungicide Use on *Aspergillus fumigatus* Abundance, Antifungal Susceptibility, and Population Structure. *mBio*. 2020. ^{*}corresponding authors

Full publication list available at: <https://scholar.google.com/citations?user=COgbwJ4AAAAAJ>

Funding

2025 - 2030	Wellcome Trust. Flipping the switch on fungal pH adaptation. <i>Role: Co-PI</i> .
2025 - 2027	CIFAR Azrieli Global Scholar Program.
2025 - 2027	Boehringer Ingelheim Foundation. Environmental interactions shape human virulence in fungi.
2025	JSMC. Mining Metagenome-Assembled Genomes for Novel Antifungal Compounds.
2025	LIFE Connect. Radically different: heterogeneity in fungal oxidative stress resistance impacts interactions with host immune cells. <i>Role: Co-PI</i> .
2024 - 2027	Leibniz Competition. Fungal RNA Transmission Impacting Human Epigenome Regulation. <i>Role: Co-PI</i> .
2024 - 2026	ProChance ^{career} . Functionalizing the pangenome of <i>Aspergillus fumigatus</i> .
2024	Balance of the Microverse Flexible Funds.
2023 - 2024	FSU Stipendium for Female Postdocs. Characterization of long non-coding RNAs associated with <i>Aspergillus</i> spp. virulence. <i>Role: Host</i>
2023	Balance of the Microverse Flexible Funds.
Since 2020	Junior research group funded by the DFG Excellence Cluster Balance of the Microverse. UK MRC Early Career Fellowship declined.
2019 - 2022	Contributed to FINAR 2.0 funded by the BMBF/InfectControl.
2012 - 2014	T-32 Microbial Pathogenesis Doctoral Training Grant funded by NIAID.

Honors and Awards

2026	Short talk winner, Wellcome Advances in Infection and Respiratory Immunology
2025	Main research prize, German speaking Mycological Society
2025-27	CIFAR Azrieli Global Scholar
2022	Publication Prize, German speaking Mycological Society
2021-22	Leibniz Mentoring Program
2021	Medac Research Prize
2018	International Society for Human and Animal Mycology Attendance Grant
2017	European Confederation of Medical Mycology Young Investigator Award
2015	ASM Richard and Mary Finkelstein Travel Award
2015	University of Utah Graduate School Travel Award
2012	University of Utah Bioscience Symposium Poster

Trainees Supervised

Since 2021	Postdocs: <i>L. Fabre, F. Palmieri, Y. Tang</i> PhD students: <i>P. Brassington, A. Pereira, M. Perrie, N. Ang</i> Masters students: <i>H. Lin, L. Forslund</i> Bachelors students: <i>P. Geiberger, L. Krämer</i> Guest researchers: <i>A. Ryan, J. Adeoye, S. Cendon, D. Rongchen</i>
2012 - 2022	Co-supervision of: PhD students: <i>T. Sae-Ong, D. Loos</i> // Masters student: <i>A. Lange</i> // Bachelors students: <i>A. Tran, K. Nelson</i>

Teaching

2022 - present	~30 contacts hours annually giving lectures and co-organizing a practical module for the MSc in Microbiology program, Friedrich Schiller University
2025	Invited lecture, Junior Society for Biochemistry and Molecular Biology Jena
2024	Invited keynote, Medical Mycology course run by Uppsala University and Swedish University of Agricultural Sciences
2014 - 2015	Instructor, Initiative to Bring Science Programs to the Incarcerated (INSPIRE) Utah State Prison and Salt Lake County Jail
2014	Instructor, Mechanisms of Bacterial Pathogenesis, University of Utah
2013	Teaching Assistant, Mechanisms of Bacterial Pathogenesis, University of Utah
2011	Teaching Assistant, Introduction to Medical Microbiology, University of Utah

Academic Service and Society Memberships

- Peer review for mBio, PLOS Pathogens, Nature Microbiology, Current Biology, PNAS, Nucleic Acids Research, Microbial Genomics, G3, Antimicrobial Agents and Chemotherapy, Environmental Microbiology, Applied and Environmental Microbiology, Communications Biology
- Guest editor at PLOS Genetics
- External grant reviewer for UKRI (UK) and BSF (Israel)
- Search committee member for three professor or junior group leader appointments at Uni Jena

Since 2023	Elected member, Aspergillus Genomes Research Policy Committee, organizing committee for Asperfest
Since 2023	Member, Association for General and Applied Microbiology (VAAM)
Since 2021	Member, German speaking Mycological Society (DMykG)
2024	Local organizer, Myk2024 conference
2020 - 2024	National Research Data Infrastructure NFDI4Microbiota representative for the Microverse Cluster and the Leibniz-HKI
2022 - 2022	Invited member, Scientific Network for Antifungal Resistance (SN-AFR)
2020 - 2022	Support CRC/Transregio 124 FungiNet with infrastructure subproject
2018- 2020	Member, International Society for Human and Animal Mycology (ISHAM)
2013 - 2016	Member, American Society of Microbiology (ASM)
2012 - 2015	University of Utah Academic Misconduct Committee in Graduate Biosciences
2011	University of Utah Department of Pathology Faculty Retention and Promotion Committee

Invited Presentations

2026	Institute Pasteur. <i>Paris, FR.</i>
2026	Antimicrobial Resistance – Genomes, Big Data and Emerging Technologies. <i>Hinxton, UK.</i>
2026	Fungal Genetics Conference. <i>Pacific Grove, CA, USA.</i>
2025	University of Guelph. <i>Guelph, CA.</i>
2025	University of Toronto. <i>Toronto, CA.</i>
2025.	Eurofung. <i>Göttingen, DE.</i>
2025	BMFTR Infection Biology Junior Group Meeting. <i>Jena, DE.</i>
2025	University of Wisconsin-Madison. <i>Madison, WI, USA</i>
2025	ESCMID, joint webinar of ESGEM and EFISG study groups. <i>Virtual.</i>
2025	Annual Meeting, Microbiology Society. <i>Liverpool, UK</i>
2025	University of Manchester. <i>Manchester, UK.</i>
2024	University of Exeter. <i>Exeter, UK.</i>
2024	GRC Cellular and Molecular Fungal Biology. <i>Holderness, NH, USA.</i>
2024	British Society for Medical Mycology Annual Meeting. <i>Cardiff, UK.</i>
2024	FEBS Host-Pathogen Interactions in Human Fungal Pathogens. <i>La Colle-sur-Loup, FR.</i>
2023	University of Exeter, Centre for Medical Mycology MycoTalks. <i>Virtual.</i>
2023	University of Neuchâtel. <i>Neuchâtel, CH.</i>
2023	University of Gießen. <i>Gießen, DE.</i>
2023	Asperfest, European Conference on Fungal Genetics. <i>Innsbruck, AT.</i>
2022	Antifungal Resistance: An Environmental Update. <i>Wageningen, Netherlands.</i>
2022	Microbial Communities: Current Approaches and Open Challenges. <i>Cambridge, UK.</i>
2022	University of Basel Biozentrum. <i>Basel, CH.</i>
2022	DMykG Mykosen am Mittwoch. <i>Virtual.</i>

Other Selected Presentations

2026	AIR 2026: Advances in Infection and Respiratory Immunology. <i>Hinxton, UK</i>
2025	European Conference on Fungal Genetics. <i>Dublin, IE.</i>
2024	Scientific Conference of the German-speaking Mycological Society (DMykG). <i>Jena, DE.</i>
2024	Joint Conference of the DGHM/VAAM. <i>Würzburg, DE.</i>
2023	Jena Microbiome Meeting. <i>Jena, DE.</i>
2023	Reinhardtsbrunn Symposium. <i>Freidrichroda, DE.</i>
2023	European Conference on Fungal Genetics. Main meeting & Fusarium Workshop. <i>Innsbruck, AT.</i>
2022	Global Health Institute, Swiss Federal Institute of Technology. <i>Lausanne, Switzerland.</i>
2022	FEBS Host-Pathogen Interactions in Human Fungal Pathogens. <i>La Colle-sur-Loup, France.</i>
2022	EMBL The Next Generation of Infection Biology. <i>Virtual.</i>
2022	Advances Against Aspergillosis and Mucormycosis. <i>Virtual.</i>
2021	Scientific Conference of the DMykG. <i>Virtual.</i>
2020	European Conference on Fungal Genetics. <i>Rome, IT.</i>
2019	Trends in Medical Mycology Conference. <i>Nice, FR.</i>
2019	Fungal Genetics Conference. <i>Pacific Grove, CA, USA.</i>
2017	Visions for Infection Control: InfectControl Strategy Conference. <i>Jena, DE.</i>
2015	ASM Microbe. <i>New Orleans, LA, USA.</i>
2012	Institute of Biology Education. University of Wisconsin-Madison. <i>Madison, WI, USA</i>

Outreach

2019	Long Night of Science Jena
2019	Skype a Scientist
2018	World Sepsis Day
2017	Long Night of Science Jena
2014	McMillian Elementary School
2014	University of Utah Educational Outreach Program
2012 – 2013	Explore Utah Science